

Summar AI

v3 Ideas — Roadmap of features for the ultimate meta learning platform

The shift for v3: from “AI writes about your materials” → “AI **plus interactive artifacts**, tailored to the subject and the type of problem.” Below is the catalog of ideas, organized into pillars.

PILLAR 1

Universal problem-solving spine

The backbone that applies across every subject. Detect what kind of problem is in the material, route it through a solver, keep every claim traceable to its source.

- **Auto task solver** — detect problem-style content (equations, prompts, “solve X”), route through per-domain logic. Every solved task gets a step-by-step explanation, a drawing when relevant, and a citation footer showing exactly which page of which PDF (or which web source) supplied each claim.
- **Web search with citations** — the model can search when your materials don't cover something. Every used source rendered as a numbered footnote you can click to open. Currently 100% closed-materials mode.
- **Progress dashboard** — per-subject mastery %, time invested, streak, “next up” recommendation.
- **Weak spot detection** — after a quiz, detect which topics you got wrong and auto-generate a targeted drill for those specific gaps.

PILLAR 2

Subject-aware interactive artifacts

Instead of returning only HTML text, the model picks a **widget type** for the situation. A registry of React components covers the domains below. This is the most differentiated capability.

Geography

- Click-the-map quiz (world / continent SVGs)
- Capitals drill, flag identification, drag-borders
- “Locate this city” on a blank map

Chemistry

- 3D molecule viewer (SMILES → rotatable model)
- Reaction balancer with instant validation
- pH / titration slider, energy diagrams

Physics

- Playable simulations with sliders (projectile with v_0 + angle, spring, wave superposition)

- Free-body diagram builder
- Circuit builder with live current / voltage readouts

Math

- Live function plotter (adjust parameters, watch the graph change)
- Geometry construction canvas (rulers, compass, angle)
- Matrix operations, complex number planes

Programming

- In-browser code editor + sandboxed test runner (Monaco + web workers)
- Support for JS, Python via Pyodide, more later
- Auto-grade against test cases, hint on failure

Biology

- Anatomy labeling (drag labels onto SVG body / cell parts)
- Phylogenetic tree walker
- Punnett square generator, DNA transcription tool

History

- Draggable timeline (drop events into order)
- Animated maps of empires over time
- Cause-and-effect graph builder

Music theory

- Interactive staff — click a note, hear it
- Chord / scale trainer with audio
- Interval ear-training

Languages

- TTS pronunciation with comparison to your recording
- Dictation drills, verb-conjugation quizzes
- Sentence-building with drag pieces

PILLAR 3

Study techniques as first-class modes

Research-backed learning methods that most study apps ignore. Each becomes its own mode in the app.

Spaced repetition (Anki-style)

Flashcards auto-generated from your materials. Daily review widget shows “8 cards today.” Mastery decays over time; weak cards resurface more often.

Feynman mode

You explain a concept in your own words. The AI plays a confused student and asks “wait, why does that work?” — catches gaps in your understanding.

Blurting

Dump everything you remember about topic X into a text box. The AI compares against your materials and shows exactly what you missed.

Retrieval practice

Hide the guide. Quiz yourself by topic name. Reveal after answering. Proven far more effective than passive re-reading.

Interleaved practice

Mix problems from different topics in a single session — research shows this beats block practice for long-term retention.

PILLAR 4

Exam prep specifically

- **Exam predictor** — analyze uploaded past exams, predict likely questions and topics for this year with reasoning.
- **Timed practice exam generator** — mimics the format of your uploaded exams, timer, auto-grade at the end.
- **Written answer grading** — you write, the AI grades against rubric criteria and suggests concrete improvements.
- **Weak-spot heat map** — a visual grid of topics color-coded by your quiz performance so you know exactly where to focus.

PILLAR 5

Voice + accessibility

- **Voice mode** — talk to the AI on a walk. Uses the browser's Web Speech API (free).
- **TTS reading** — have any study guide or section read aloud.
- **Handwritten note capture** — take a photo of your notes, OCR extracts text, integrated as a material.

PILLAR 6

Infrastructure v3 will need

Cost / effort estimate for the underlying platform pieces.

Component	Purpose	Cost / effort
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Web search API	Cite external sources when materials don't cover something	Tavily (~\$10/mo) / Exa / Perplexity
Per-user database	Replace localStorage; sync across devices; enable sharing and spaced repetition schedules	Supabase / Vercel Postgres — free tier available
Artifact registry	Set of React widgets the model can invoke via a JSON schema	Build in-house per widget
Image generation	Optional: illustrations for concepts that need them	DALL-E, Flux, Stability — pay per image

Signature v3 picks

If v3 shipped only these five, it would already be the most differentiated learning tool on the market. Most competitors do one of these; nobody does all five well.

1. Interactive artifacts

Map-click geography, physics simulations, math plotters, molecule viewers, code runners — generated on demand per problem. This is what turns Summar AI from “a chatbot with materials” into “a real learning environment.”

2. Automatic exam solver with citations

Upload any past exam PDF. Every single question gets solved with full step-by-step reasoning, drawings when relevant, and citations back to your materials or the web. This alone justifies the whole platform for exam season.

3. Spaced repetition backed by your materials

Anki, but the cards come from your own PDFs and lectures — no separate deck-building workflow. Review widget on the home page: “8 cards due today.”

4. Feynman + Blurting + Retrieval modes

Study techniques that actually work, packaged as first-class app modes with AI feedback. Essentially nobody else offers these properly.

5. Web search with citations

Closes the “materials don’t cover it” gap while staying trustworthy — every external claim gets a clickable numbered footnote to the source.

Next step: pick a pillar (or a signature feature) to draft an actual v3 architecture — component registry shape, DB schema, API surface. Not pushing anything until we’re ready.